arXiv:2305.18474v1 [cs.SD] 29 May 2023

Make-An-Audio 2: Temporal-Enhanced Text-to-Audio Generation

Jiawei Huang[*†]
Zhejiang University, ByteDance
huangjw@zju.edu.cn

Yi Ren[*]
ByteDance
ren.yi@bytedance.com

Rongjie Huang
Zhejiang University
rongjiehuang@zju.edu.cn

Dongchao Yang
Peking University
2001212832@stu.pku.edu.cn

Zhenhui Ye[†]
Zhejiang University, ByteDance
zhenhuiye@zju.edu.cn

Chen Zhang, Jinglin Liu, Xiang Yin, Zejun Ma
ByteDance
{zhangchen.0620,jinglinliu,yinxiang.stephen,mazejun}@bytedance.com

Zhou Zhao[‡]
Zhejiang University
zhaozhou@zju.edu.cn

Abstract

Large diffusion models have been successful in text-to-audio (T2A) synthesis tasks, but they often suffer from common issues such as semantic misalignment and poor temporal consistency due to limited natural language understanding and data scarcity. Additionally, 2D spatial structures widely used in T2A works lead to unsatisfactory audio quality when generating variable-length audio samples since they do not adequately prioritize temporal information. To address these challenges, we propose Make-an-Audio 2, a latent diffusion-based T2A method that builds on the success of Make-an-Audio. Our approach includes several techniques to improve semantic alignment and temporal consistency: Firstly, we use pre-trained large language models (LLMs) to parse the text into structured <event & order> pairs for better temporal information capture. We also introduce another structured-text encoder to aid in learning semantic alignment during the diffusion denoising process. To improve the performance of variable length generation and enhance the temporal information extraction, we design a feed-forward Transformer-based diffusion denoiser. Finally, we use LLMs to augment and transform a large amount of audio-label data into audio-text datasets to alleviate the problem of scarcity of temporal data. Extensive experiments show that our method outperforms baseline models in both objective and subjective metrics, and achieves significant gains in temporal information understanding, semantic consistency, and sound quality.

[*]Equal contribution.
[†]Interns at ByteDance.
[‡]Corresponding author

Preprint. Under review.

1 Introduction

Deep generative learning models [9, 22, 12] have revolutionized the creation of digital content, enabling creators with no professional training to produce high-quality images [44, 45, 36], vivid videos [14, 50], diverse styles of voice [17], and meaningful long textual spans [57, 37]. Professional practitioners can modify the generated content to accelerate their production workflows. Text-to-audio synthesis (T2A) is a subcategory of generative tasks that aims to generate natural and accurate audio by taking text prompts as input. T2A can be useful in generating desired sound effects, music or speech, and can be applied to various applications like movie sound effects making, virtual reality, game development, and audio editing.

Thanks to the development of text-to-image synthesis (T2I) methods, researchers have successfully extended similar approaches to the text-to-audio synthesis domain [15, 29, 55, 25]. The success of these methods has opened up numerous opportunities for generating high-quality audio content from text. T2A systems typically use a text encoder to encode the audio's text input as condition embedding, then employ diffusion models [15, 29, 55] to synthesis mel-spectrograms or utilize autoregressive models [25] to synthesis raw waveform data based on the condition embedding. However, previous T2A methods have some common issues: 1) **Temporal disorder**: when the text input is complex, with multiple objects and temporal relationships between them, the generated audios often suffer from semantic misalignment and temporal disorder. For instance, audio captions such as "The sound of A, followed by the sound of B" may result in audios where A and B overlapping throughout, or B comes before A, or even only one sound is synthesized. 2) **Poor variable-length results**: previous works [15] adopt the conventional U-Net structure of 2D convolution and spatial transformer stacking as the backbone of diffusion denoiser, which is typically trained with fixed-length audios. Consequently, they generate suboptimal results when synthesizing audio sequences of varying lengths compared to those of the training data. In the meanwhile, 2D spatial structures are not good at extracting temporal information since they treat the time axis and frequency axis equally in the spectrogram generation process. 3) **Insufficient temporal paired data**: previous works use simple rule-based augmentation methods [6, 25] to create temporally aligned text-audio paired data from audio-label datasets. However, these patterns are overly simplistic and can hinder the model's ability to generalize to real-world sentences.

In this paper, based on previous successful work Make-An-Audio, we propose a novel temporal-enhanced text-to-audio generation framework called Make-An-Audio 2. The temporal information can be better handled by our method in the following ways: 1) To address the semantic misalignment and temporal disorder, we use a pre-trained LLM to extract the audio caption's temporal information and parse the origin caption into structured <event & order> pairs with proper prompts. To encode the structured pairs better, we introduce another structured-text encoder that takes the structured pairs as its input to aid in learning semantic alignment during the diffusion denoising process. In this way, we relieve the text encoder's burden of recognizing events with the corresponding temporal information and enable the T2A system to model the timing information of the events more effectively. 2) To improve the generation quality of variable-length audio and enhance the temporal information understanding, we replace the 2D spatial structures with temporal feed-forward Transformer [53] and 1D-convolution stacks for the diffusion denoiser and support variable-length audio input in training. 3) To address the issue of insufficient temporally aligned audio-text paired dataset, we use single-labeled audio samples and their labels to compose complex audio and structured captions. We then use LLM to augment the structured caption into natural language captions.

We conduct extensive experiments on AudioCaps and Clotho datasets, which reveals that our method surpasses baseline models in both objective and subjective metrics, and achieves significant gains in understanding temporal information, maintaining semantic consistency, and enhancing sound quality. Our ablation studies further demonstrate the effectiveness of each of our techniques[4].

[4]Our demos are available at `https://make-an-audio-2.github.io/`

2 Related works

2.1 Text-to-image generative models

Text-to-Image Synthesis (T2I) has garnered significant attention in recent years and has even been commercialized. One pioneering work in this realm is DALL-E [43], which treats T2I generation as a sequence-to-sequence translation task. DALL-E employs a pre-trained VQ-VAE [52] to encode image patches to discrete codes, which are then combined with the text codes. During inference, the model generates image codes autoregressively based on the text codes. As diffusion models exhibit greater potential with regard to both diversity and quality in image generation, they have become mainstream in T2I Synthesis. DALLE-2 [42] uses the CLIP [40] text encoder and two diffusion models. The first diffusion model predicts CLIP visual features based on the CLIP text feature, while the second synthesizes the image from the predicted CLIP visual features. A cascade of diffusion super-resolution models is then employed to increase the resolution of the generated image. Another famous T2I work is Imagen [45], which utilizes the T5 encoder [41] to extract text features, It employs a diffusion model to synthesize a low-resolution image and then applies a cascade of diffusion models for super-resolution. Latent Diffusion [44] enhances computational efficiency by using a continuous VAE that is trained with a discriminator to map images from pixel space to compressed latent space. This is followed by diffusion on the latent space, which synthesizes images' latent.

2.2 Text-to-audio synthesis

Text-to-Audio Synthesis is a rising task that has seen great advances recently. Diffsound [55] uses a pre-trained VQ-VAE [52] trained on mel-spectrograms to convert audio into discrete codes, which are then used by a diffusion model to generate the audio codes. To improve its generalization ability, the authors pre-trained Diffsound on the AudioSet dataset, which contains audio files labeled with tags. Additionally, they introduce a random input masking technique to make use of these tags. AudioGen [25] is another system in this field that uses a similar VQ-VAE-based approach. However, it encodes raw waveform data into discrete codes and employs an autoregressive model to predict audio tokens based on text features. For data augmentation, AudioGen mixes audio files and concatenates their text captions. Make-An-Audio [15], AudioLDM [29], and TANGO [8] are all based on the Latent Diffusion Model (LDM). With the assumption that CLAP can map the audio and its caption to the same latent space and approximate the text features based on the audio feature, AudioLDM uses audio features extracted by the CLAP model as the condition during training and utilizes text features during inference. In contrast, Make-An-Audio and TANGO employ text features both in the training and inference stages. To overcome data scarcity, Make-An-Audio proposes a pseudo prompt enhancement method, while TANGO introduces an audio mixture method based on human auditory perception.

2.3 LLM-based data augmentation

Recent advancements in prompt learning [30] have greatly enhanced the capabilities of language models and given birth to very large language models with billions of parameters, enabling them to achieve natural language comprehension levels that are comparable to those of humans. OPT [57], ChatGPT[1] and GPT4 [37], are typical cases among them. This has led researchers to explore whether these models can be used to annotate data instead of humans. In previous work, masked language models (MLM) like BERT [4] and Roberta [31] have been used for contextual augmentation [27] at the word level. For example, researchers insert <mask> tokens into the text or replace some words with <mask> tokens, and then use the MLM to predict the appropriate words. At the sentence level, back translation [48] and paraphrasing [26] methods have been used to increase the diversity of data. However, the limited capabilities of these models have resulted in insufficient quality and diversity of the generated data. To address these limitations, recent research has explored the use of very large language models for data augmentation. AugGPT [2], for instance, leverages ChatGPT to generate auxiliary samples for few-shot text classification. The quality of the generated data is much higher, resulting in double-digit improvements in sentence classification accuracy compared to previous data augmentation methods. WavCaps [33] crawls audio data with raw descriptions from multiple web

[1] `https://openai.com/blog/chatgpt`

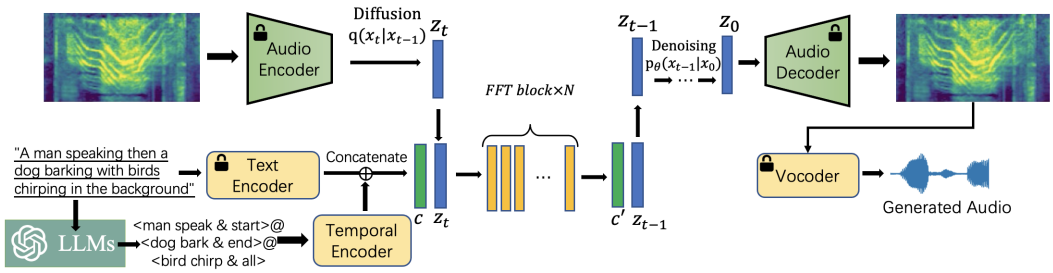

Figure 1: A high-level overview of Make-An-Audio 2. Note that modules printed with a *lock* are frozen when training the T2A model.

sources. However, the raw descriptions contain a high degree of noise, so ChatGPT is used to filter out extraneous information unrelated to the audio and generate high-quality captions based on labels. This approach results in a large-scale, weakly-labeled audio captioning dataset.

3 Preliminary

Our work is inspired by Make-An-Audio [15], which is one of the most successful T2A models. In this section, we provide a preliminary discussion on Make-An-Audio. Denote an audio-text pair as (a, y) where $a \in R^{T_a}$ and T_a is the waveform length. To mitigate the complexity of modeling long continuous waveform data, they first convert a to mel-spectrogram (akin to 1-channel 2D images) $x \in R^{C_a \times T}$, where $C_a, T \ll T^a$ denote the mel-channels and the number of frames respectively. The training process includes two stages:

1) **Training variational autoencoder**. The audio encoder E takes mel-spectrogram x as input and outputs compressed latent $z = E(x)$. The audio decoder D reconstructs the mel-spectrogram signals $x' = D(z)$ from the compressed representation z. VAE solves the problem of excessive smoothing in mel-spectrogram reconstruction through adversarial training with a discriminator. The training objective is to minimize the weighted sum of reconstruction loss $\mathcal{L}_{re}$, GAN loss $\mathcal{L}_{GAN}$ and KL-penalty loss $\mathcal{L}_{KL}$.

2) **Training latent diffusion model**. The text encoder f_{text} encodes the text input y into conditional embedding $c = f_{text}(y)$. The training objective of the U-Net diffusion module is to minimize the mean squared error in the noise space:

$$\mathcal{L}_\theta = \|\epsilon_\theta(z_t, t, c) - \epsilon\|_2^2, \tag{1}$$

where, $\epsilon \sim \mathcal{N}(0, I)$ denotes the noise, ϵ_θ denotes the denoising network, t is the random time step, c is involved by the cross-attention mechanism. The diffusion model can be efficiently trained by optimizing ELBO, ensuring extremely faithful reconstructions that match the ground-truth distribution.

During inference, with the conditional embedding c from text and the noise z_t sampled from Gaussian distribution, the diffusion network runs denoising steps to get z_0. By applying the audio decoder D and the separately trained Vocoder V, the generated audio is got by $a' = V(D(z_0))$. To enhance control over conditional generation, Make-An-Audio applies classifier-free guidance [13] technique:

$$\tilde{\epsilon}_\theta(z_t, t, c) = \epsilon_\theta(z_t, t, c_\emptyset) + s \cdot (\epsilon_\theta(z_t, t, c) - \epsilon_\theta(z_t, t, c_\emptyset)), \tag{2}$$

where $c_\emptyset$ denotes the conditional embedding when empty prompt is given, s is the guidance scale. The model reduces the probability of generating samples that do not use conditioning information, in favor of the samples that explicitly do as s increases.

Pseudo prompt enhancement method with audio captioning model [54] and audio-text retrieval model [23] is also introduced to construct captions for audios without natural language annotation. However, it should be noted that the quality of the captions generated by the audio captioning model and the diversity of the caption templates are limited.

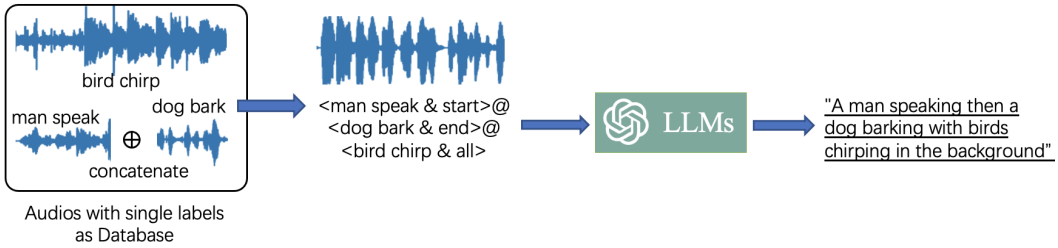

Figure 2: Overview of LLM-based data augmentation. We use single-labeled audios and their labels as a database. Composing complex audios and the structured captions with these data. We then use LLM to generate diverse natural language captions by the constructed captions and appropriate prompt.

4 Make-An-Audio 2

In this section, we first describe the motivation of several designs in Make-An-Audio 2, and then introduce the overall architecture as illustrated in Figure 1, to enhance temporal modeling in text-to-audio generation.

4.1 Motivation

In comparison to image data, audio data includes temporal information. A sound event can occur at any time within the audio, making audio synthesis a challenge when attempting to maintain temporal consistency. Previous approaches have encountered difficulties in dealing with captions that contain multiple sounds and complex temporal information, leading to semantic misalignment and poor temporal consistency. This can cause the generated audio to omit some sounds and produce an inaccurate temporal sequence. To address these issues, we propose the **temporal enhancement** method by parsing the original caption into structured pairs of <event & order>.

Additionally, we introduce a temporal encoder that utilizes the structured caption to help the model better understand the temporal information regarding the event sound. The temporal encoder and the main text decoder form our **dual text encoders** architecture. Another challenge in T2A synthesis pertains to data scarcity. Most of the data in this task is "dirty", which means there are extra noises and other sounds in the audio aside from the annotated sounds. Furthermore, there is little data available with detailed temporal annotation. To combat this issue, we propose a structured data construction and **LLM-based data augmentation** approach.

Previous diffusion-based T2A work continues the T2I diffusion model architecture using 2D-convolution and spatial transformer stacked U-Net networks as the diffusion model backbone. However, mel-spectrogram, unlike images, is not spatially translational invariant and the previous architecture is not suitable for training variable-length audio data, so we propose a feed-forward **Transformer-based diffusion denoiser backbone** and 1D-convolution-based audio VAE to improve the robustness of the model for generating variable-length audio data.

4.2 Temporal enhancement

Recently, AudioGPT [16] and HuggingGPT [49] take LLM (e.g., ChatGPT) as a controller to invoke other AI models for expanding LLM's capacity in addressing multi-modal tasks. Conversely, we consider the possibility of utilizing the robust language understanding capabilities of LLMs to provide temporal knowledge. Thus, we introduce the proposed temporal enhancement method.

Specifically, LLMs are utilized to parse the input text (the natural language audio caption) and extract structured <event & order> pairs. As illustrated in Figure 1, LLMs simplify the original natural language caption and link each sound event to its corresponding order. Benefiting from enhanced temporal knowledge, the T2A model is empowered to identify sound events and corresponding temporal order. Appendix D contains further details on prompt design and additional examples of temporal information enhancement.

4.3 Dual text encoders

To enhance the utilization of caption information, we propose a dual text encoder architecture consisting of a main text encoder CLAP [6], and a temporal encoder T5 [41]. With contrastive multi-modal pre-training, the CLAP has achieved excellent zero-shot performance in several downstream tasks. We freeze the weights of the main text encoder and fine-tune the temporal encoder to capture information about the temporal order of various events. Denote the audio-text pair and the parsed structured caption respectively as $(a, y), y_s$, the final textual representation is expressed as:

$$c = Linear(Concat(f_{text}(y), f_{temp}(y_s))), \tag{3}$$

where f_{text} is the main text encoder and f_{temp} is the temporal encoder.

4.4 LLM-based data augmentation

The remarkable success of the GPT series has underscored the significance of data-centric artificial intelligence [56, 18], which seeks to optimize the creation, selection, and maintenance of training and inference data for optimal outcomes. While the T2I task benefits from billions of text-image pairs [47], there are currently only around one million open-source text-audio pairs avaiable [15]. Furthermore, many of these audios are only roughly labeled with tags instead of natural language captions. To optimize the use of this data, we propose an LLM-based data augmentation technique. As shown in Figure 2, we augment audio data and its corresponding text caption as follows:

- We begin by collecting data labeled with single tags to create our event database $\mathcal{D}$. This type of data is typically cleaner and less likely to contain unwanted noise or other sounds. We can then use this data to construct more complex data based on their durations.
- Then we randomly select $N \in \{2, 3\}$ samples from $\mathcal{D}$, mix and concatenate them at random. Concatenating at random intervals or overlaps ensures that the resulting audio contains temporal information. Mixing improves the models' ability to recognize and separate different sorts of audio for creating complex compositions.
- As the resulting audio is created, we synthesize structured captions based on the occurrence time and duration of each sound event by rules. For those events that appear almost throughout the audio, we bind them with "all". While for events that only partly occur in the audio, we bind them with "start", "mid" or "end" depending on the proportion of their occurrence time points in the resulting audio.
- Finally, we feed the structured captions into LLM with appropriate prompts to generate diverse natural language captions. The prompt to transform structured captions to natural language captions and some examples are displayed in Appendix D.

4.5 Transformer-based diffusion denoiser backbone

Previous diffusion-based work on T2A synthesis treated the mel-spectrogram as a one-channel image similar to the approach used for T2I synthesis. However, unlike images, the mel-spectrogram is not spatially translation invariant. The height of the mel-spectrogram represents the frequency domain, which means that mel-spectrogram patches at different heights can have entirely different meanings and should not be treated equally. Furthermore, the use of a 2D-convolution layer and spatial transformer-stacked U-Net architecture limits the model's ability to generate variable-length audio. Previous works [38, 1] has shown U-Net is not necessary for diffusion network [12, 44] and found transformer-based [53] architecture as diffusion network can achieve better performance. Inspired by these works and to improve the model's ability to generate variable-length audio, we propose a modified audio VAE that uses a 1D-convolution-based model and propose a feed-forward Transformer-based diffusion denoiser backbone. While the latent of Make-An-Audio's audio encoder is $z = E(x) \in R^{c_e \times C_a/f \times T/f}$, where c_e is the embedding dim of latent, f is the downsampling rate, C_a and T denote the mel-channels and the number of frames of mel-spectrogram respectively, which can be seen as images' height and width. Our 1D-convolution-based audio encoder's latent is $z = E(x) \in R^{C_a/f_1 \times T/f_2}$, where f_1, f_2 are downsampling rates of mel-channels and frames, respectively. As the feed-forward Transformer block is composed of 1D-convolution and temporal transformer, it can better understand the information of the temporal domain in latent and improves the variable length audio generation performance. Compared to the original spatial transformer, the

computation complexity reduces from $O((C_a/f \times T/f)^2 \times D)$ to $O((T/f_2)^2 \times D)$, where D is the embedding dimension of the transformer layer.

5 Experiments

5.1 Experimental setup

Dataset. We use a combination of several datasets to train our model, including: AudioCaps training set, WavCaps, AudioSet, Adobe Audition Sound Effects, Audiostock, ESC-50, FSD50K, MACS, Epidemic Sound, UrbanSound8K, WavText5Ks, TUT acoustic scene. This results in a dataset composed of **0.92 million** audio text pairs, with a total duration of approximately **3.7K hours**.

We conduct preprocessing on both text and audio data as follows: 1) We convert the sampling rate of audios to 16kHz. Prior works [55, 15, 29] pad or truncate the audio to a fixed length (10s), while we group audio files with similar durations together to form batches to avoid excessive padding which could potentially impair model performance and slow down the training speed. This approach also allows for improved variable-length generation performance. We truncate any audio file that exceeds 20 seconds, in order to speed up the training process. 2) We adopt the LLM-based data augmentation method in section 4.4 to construct approximately 61k additional audio-text pairs as auxiliary data. 3) For audios without natural language annotation, we apply the pseudo prompt enhancement method from Make-An-Audio [15] to construct captions aligned with the audio. 4) We assign a lower weightage to the data that is not annotated with temporal information, but is abundant in quantity and diversity, such as AudioSet and WavCaps data. Specifically, we traverse the AudioCaps training set and the LLM augmented data with a probability of 50%, while randomly selecting data from all other sources with a probability of 50%. For the latter dataset, we use "<text & all>" as their structured caption. To evaluate the performance of our models, we use the AudioCaps test set and Clotho evaluation set which contain multiple event audio samples and detailed audio captions. The latter serves as a more challenging zero-shot scenario test for us. More details about datasets are put in Appendix A.

Model configurations. We train a continuous 1D-convolution-based VAE to compress the mel-spectrogram to a 20-channel latent with temporal axis downsample rates of 2. For our main experiments, we train a feed-forward Transformer-based diffusion backbone. The diffusion model is trained on 8 NVIDIA A100 GPU with 1.2M optimization steps, 4 batch size per GPU. We use the AdamW optimizer [21] with a learning rate of 9.6e-5. BigVGAN [28], a universal vocoder that generalizes well for various scenarios, is used as our vocoder, we trained it on the AudioSet Dataset. For more details about Model configuration please refer Appendix B.

Evaluation methods. We evaluate our models using objective and subjective metrics to assess the audio quality and text-audio alignment faithfulness. For objective evaluation, we include Frechet distance (FD), inception score (IS), Kullback–Leibler (KL) divergence, Frechet audio distance (FAD), and CLAP score. For subjective evaluation, we conduct crowd-sourced human evaluations with MOS (mean opinion score) to assess the audio quality and the text-audio alignment faithfulness, scoring MOS-Q and MOS-F, respectively. More information regarding the evaluation process can be found in Appendix C.2

Baseline models. To establish a standard for comparison, our study employs four baseline models, including Make-An-Audio [15], AudioLDM [29], TANGO [8] and AudioGen [25]. We reimplement Make-An-Audio and train it on AudioCaps Dataset. AudioLDM-S and AudioLDM-L with 454M and 1.01B parameters respectively are trained on AudioCaps, AudioSet, BBC Sound Effects, and the FreeSound dataset. TANGO is trained on AudioCaps Dataset. The close-source model AudioGen-Base and AudioGen-Large, with 285M and 1M parameters are trained on AudioCaps, AudioSet, and eight other datasets.

5.2 Main results

Automatic objective evaluation. The objective evaluation comparison with baseline is presented in Table 1, and we have the following observations: 1) In terms of audio quality, Make-An-Audio 2 achieves better scores in FD, IS, KL, and FAD; 2) On text-audio similarity, Make-An-Audio 2

Table 1: The comparison between Make-An-Audio 2 and baseline T2A models on the AudioCaps dataset. All the diffusion-based models run with 100 DDIM [51] steps for a fair comparison. We borrowed all the results from [29, 8] and used the model released by the authors on Huggingface to test CLAP Score. We reimplement Make-An-Audio and replace their vocoder with our BigVGAN vocoder.

Model	Prams	FD↓	IS↑	KL↓	FAD↓	CLAP↑	MOS-Q↑	MOS-F↑
GroundTruth	-	-	-	-	-	0.671	86.47	84.31
AudioGen-S	285M	-	-	2.09	3.13	-	-	-
AudioGen-L	1B	-	-	1.69	1.82	-	-	-
Make-An-Audio	453M	18.32	7.29	1.61	2.66	0.593	69.54	65.45
AudioLDM-S	454M	29.48	6.9	1.97	2.43	-	-	-
AudioLDM-L	1.01B	23.31	8.13	1.59	1.96	0.605	70.91	67.41
TANGO	1.21B	26.13	8.23	1.37	1.87	**0.650**	74.10	72.76
Make-An-Audio 2	937M	**11.75**	**11.16**	**1.32**	**1.80**	0.645	**78.31**	**75.63**

presents the comparable score in CLAP; 3) Regarding temporal alignment, Make-An-Audio 2 also achieves a high IS score, showing that temporal enhancement method also enhances the clarity and expressiveness of the sound events in the generated audio.

Subjective human evaluation. The human evaluation results show significant gains of Make-An-Audio 2 with MOS-Q of 78.31 and MOS-F of 75.63, outperforming TANGO in the MOS-F evaluation. It indicates that raters prefer our model synthesis against baselines in terms of audio naturalness and faithfulness.

Zero-shot evaluation. To further investigate the generalization performance of the models, we additionally test the performance of the models on the Clotho-evaluation dataset in the zero-shot scenario. As illustrated in Table 2, Make-An-Audio 2 has significantly better results than TANGO and AudioLDM-L, attributing to the scalability in terms of data usage.

Table 2: Comparison of Make-An-Audio 2, AudioLDM-L, and Tango on Clotho-eval (10s) and AudioCaps-test (5s) datasets. We truncate the ground truth audios from AudioCaps dataset to 5 seconds to conduct variable-length generation evaluation.

Model	Clotho-eval				AudioCaps-test			
	FD↓	IS↑	KL↓	FAD↓	FD↓	IS↑	KL↓	FAD↓
TANGO	32.1	6.77	2.59	3.61	31.76	5.50	2.04	10.53
AudioLDM-L	28.15	6.55	2.6	4.93	31.97	5.66	2.39	6.79
Make-An-Audio 2	**19.97**	**8.50**	**2.49**	**2.13**	**13.78**	**9.95**	**1.61**	**2.33**

5.3 Analyses

Variable-length generation. Audio data can have different lengths, to investigate our models' performance on variable-length audio generation, we test to generate 5 seconds audios on AudioCaps dataset, the results are shown in Table 2. From the table, it can be seen that TANGO and AudioLDM exhibit significant performance degradation when generating audio with different lengths than the training data, as TANGO and AudioLDM pad or truncate all the training audio data to 10 seconds, and their models are based on 2D-convolution and spatial transformer to process mel-spectrogram as images. Make-An-Audio 2 maintains high performance even when generating variable-length audio samples since it is trained on audio samples of varying lengths and utilizes 1D-convolution and temporal transformers to emphasize temporal information.

Classifier-free guidance scale. We explore the classifier-free guidance in text-to-audio synthesis. The choice of the classifier guidance scale could scale conditional and unconditional synthesis, offering a trade-off between sample faithfulness and realism with respect to the conditioning text. We show the effect of guidance scale w on T2A in Table 3. Our results show that a guidance scale of $w = 4$ yields the best performance in terms of FD and FAD.

Table 3: Comparison of different classifier-free guidance scales on AudioCaps-test set.

Guidance Scale	FD↓	IS↑	KL↓	FAD↓
1	30.13	4.61	2.26	7.55
2	20.71	7.23	1.66	3.76
3	15.71	9.07	1.47	2.41
4	**11.75**	11.16	1.32	**1.80**
5	12.43	**11.37**	**1.25**	1.95
7.5	13.96	11.32	1.31	2.97
10	16.06	10.52	1.47	3.59

Table 4: Vocoder reconstruction performance comparison on AudioCpas-test set.

Vocoder	Model	FD↓	IS↑	KL↓	FAD↓
MelGAN	Diffsound	26.14	5.4	1.22	6.24
HifiGAN	Make-An-Audio	21.79	5.93	1.03	6.02
HifiGAN	AudioLDM	11.45	8.13	0.22	1.18
BigVGAN	Make-An-Audio 2	5.45	9.44	0.17	0.98

Vocoder performance comparison. Vocoder is another key component to improve the generated audio quality for mel-spectrogram-based models. We further compared the 1) BigVGAN vocoder used in Make-An-Audio 2, 2) AudioLDM's pre-trained HifiGAN, 3) Make-An-Audio's pre-trained HifiGAN and 4) Diffsound's pre-trained MelGAN. The results are shown in Table 4. We find 3) and 4) both perform worse compared with 2), while 2) and 3) use the same vocoder architecture. We find that the problem lies in the mel-processing method of 3) and 4), they use the same mel-processing method which will result in poor performance of the vocoder. So we adopt the same mel-spectrogram extraction process as BigVGAN and get the best performance.

5.4 Ablation study

Table 5: The results evaluated on AudioCaps-test set with different settings of text encoder, diffusion model backbone, and dataset. Noted that † marks the text encoder is frozen, * means trainable one.

Setting	Text encoder	Backbone	Dataset	FD↓	IS↑	KL↓	FAD↓
#1	CLAP†	U-Net	AudioCaps	18.24	7.46	1.61	2.25
#2	T5*	U-Net	AudioCaps	13.73	9.79	1.41	2.05
#3	CLAP†+T5*	U-Net	AudioCaps	13.71	9.99	1.36	1.92
#4	CLAP†+T5*	U-Net	Variable length All	22.69	5.93	2.17	3.82
#5	CLAP†+T5*	Transformers	Variable length All	11.75	11.16	1.32	1.80

In order to assess the effectiveness of various designs in Make-An-Audio 2 including temporal enhancement, dual text encoder, and feed-forward Transformer-based diffusion denoiser, we conduct ablation studies. The results evaluated on the AudioCaps-test set with different settings for text encoder, diffusion denoiser backbone, and dataset are presented in Table 5. The key findings are discussed below:

- Comparing setting #2 to #1 highlights the effectiveness of temporal enhancement. We use LLM to extract event and temporal information and create structured input in the form of <event & order> pairs. As this format of structured input is not in the training corpus of the text encoder, we use the trainable T5 as our text encoder, which leads to significant improvements in both objective scores and sound timing modeling.
- Comparing setting #3 with #2 proves the effectiveness of dual text encoder architecture. The reason is that the structured caption parsed by LLM may result in the loss of some information such as adjectives and quantifiers, and the trainable text encoder may reduce generalization abilities. Therefore, it is still essential to use the frozen CLAP encoder to comprehend the original natural language caption.

- For setting #4, we train our model with the large combined dataset instead of AudioCaps-train set and switch the training mode from fixed-length audio training to variable-length audio, resulting in severe performance degradation. As 2D-convolution and spatial Transformer treat the mel-spectrograms as images, and it treats the temporal information and frequency information the same way, they demonstrate deficiencies in understanding variable-length data with temporal information.
- Setting #5 is our Make-An-Audio 2. Compared with setting #4, we replace the 2D-convolutional VAE with 1D-convolutional VAE and replace the U-Net diffusion backbone with feed-forward Transformer layers and observe performance gains, which demonstrates a strong ability in understanding variable length audios and temporal information.

6 Conclusions

In this work, we present Make-An-Audio 2, a temporal-enhanced T2A synthesis model. With a capable LLM to extract temporal information from the natural language caption, Make-An-Audio 2 can better understand the event order in the caption and generate semantically aligned audios. Leveraging 1D-convolutional VAE and feed-forward Transformer diffusion backbone, Make-An-Audio 2 can generate variable-length audios without performance degeneration. With complex audio reconstruction and LLM-based data augmentation, Make-An-Audio 2 is endowed with the ability to understand complex temporal relationships and combinations of multiple concepts. Make-An-Audio 2 achieves the SOTA audio generation quality in both objective and subjective metrics, and extends the boundaries of the T2A. We discuss the limitations, future works and broader impact in Appendix E.

References

[1] Fan Bao, Shen Nie, Kaiwen Xue, Yue Cao, Chongxuan Li, Hang Su, and Jun Zhu. All are worth words: A ViT backbone for diffusion models.

[2] Haixing Dai, Zhengliang Liu, Wenxiong Liao, Xiaoke Huang, Yihan Cao, Zihao Wu, Lin Zhao, Shaochen Xu, Wei Liu, Ninghao Liu, Sheng Li, Dajiang Zhu, Hongmin Cai, Lichao Sun, Quanzheng Li, Dinggang Shen, Tianming Liu, and Xiang Li. Auggpt: Leveraging chatgpt for text data augmentation, 2023.

[3] Soham Deshmukh, Benjamin Elizalde, and Huaming Wang. Audio retrieval with WavText5K and CLAP training.

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.

[5] Konstantinos Drossos, Samuel Lipping, and Tuomas Virtanen. Clotho: An audio captioning dataset. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 736–740.

[6] Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. Clap: Learning audio concepts from natural language supervision.

[7] Jort F Gemmeke, Daniel PW Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R Channing Moore, Manoj Plakal, and Marvin Ritter. Audio set: An ontology and human-labeled dataset for audio events. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 776–780.

[8] Deepanway Ghosal, Navonil Majumder, Ambuj Mehrish, and Soujanya Poria. Text-to-audio generation using instruction-tuned llm and latent diffusion model, 2023.

[9] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. 63(11):139–144.

[10] Shawn Hershey, Sourish Chaudhuri, Daniel PW Ellis, Jort F Gemmeke, Aren Jansen, R Channing Moore, Manoj Plakal, Devin Platt, Rif A Saurous, Bryan Seybold, et al. CNN architectures for large-scale audio classification. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 131–135. IEEE.

[11] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning.

[12] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Proc. of NeurIPS*.

[13] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*.

[14] Wenyi Hong, Ming Ding, Wendi Zheng, Xinghan Liu, and Jie Tang. CogVideo: Large-scale pretraining for text-to-video generation via transformers.

[15] Rongjie Huang, Jiawei Huang, Dongchao Yang, Yi Ren, Luping Liu, Mingze Li, Zhenhui Ye, Jinglin Liu, Xiang Yin, and Zhou Zhao. Make-an-audio: Text-to-audio generation with prompt-enhanced diffusion models, 2023.

[16] Rongjie Huang, Mingze Li, Dongchao Yang, Jiatong Shi, Xuankai Chang, Zhenhui Ye, Yuning Wu, Zhiqing Hong, Jiawei Huang, Jinglin Liu, Yi Ren, Zhou Zhao, and Shinji Watanabe. Audiogpt: Understanding and generating speech, music, sound, and talking head, 2023.

[17] Rongjie Huang, Yi Ren, Jinglin Liu, Chenye Cui, and Zhou Zhao. Generspeech: Towards style transfer for generalizable out-of-domain text-to-speech. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022.

[18] Johannes Jakubik, Michael Vössing, Niklas Kühl, Jannis Walk, and Gerhard Satzger. Data-centric artificial intelligence. *arXiv preprint arXiv:2212.11854*, 2022.

[19] Kevin Kilgour, Mauricio Zuluaga, Dominik Roblek, and Matthew Sharifi. Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms. In *INTERSPEECH*, pages 2350–2354.

[20] Chris Dongjoo Kim, Byeongchang Kim, Hyunmin Lee, and Gunhee Kim. AudioCaps: Generating captions for audios in the wild. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 119–132.

[21] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint:1412.6980*, 2014.

[22] Durk P Kingma and Prafulla Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. 31:10215–10224.

[23] A Sophia Koepke, Andreea-Maria Oncescu, Joao Henriques, Zeynep Akata, and Samuel Albanie. Audio retrieval with natural language queries: A benchmark study. *IEEE Transactions on Multimedia*, 2022.

[24] Qiuqiang Kong, Yin Cao, Turab Iqbal, Yuxuan Wang, Wenwu Wang, and Mark D Plumbley. PANNs: Large-scale pretrained audio neural networks for audio pattern recognition. 28:2880–2894.

[25] Felix Kreuk, Gabriel Synnaeve, Adam Polyak, Uriel Singer, Alexandre Défossez, Jade Copet, Devi Parikh, Yaniv Taigman, and Yossi Adi. AudioGen: Textually Guided Audio Generation.

[26] Ashutosh Kumar, Satwik Bhattamishra, Manik Bhandari, and Partha Talukdar. Submodular optimization-based diverse paraphrasing and its effectiveness in data augmentation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3609–3619, 2019.

[27] Varun Kumar, Ashutosh Choudhary, and Eunah Cho. Data augmentation using pre-trained transformer models, 2021.

[28] Sang-gil Lee, Wei Ping, Boris Ginsburg, Bryan Catanzaro, and Sungroh Yoon. BigVGAN: A Universal Neural Vocoder with Large-Scale Training.

[29] Haohe Liu, Zehua Chen, Yi Yuan, Xinhao Mei, Xubo Liu, Danilo Mandic, Wenwu Wang, and Mark D. Plumbley. AudioLDM: Text-to-Audio Generation with Latent Diffusion Models.

[30] Pengfei Liu, Weizhe Yuan, Jinlan Fu, Zhengbao Jiang, Hiroaki Hayashi, and Graham Neubig. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing, 2021.

[31] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

[32] Irene Martín-Morató and Annamaria Mesaros. What is the ground truth? reliability of multi-annotator data for audio tagging. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 76–80.

[33] Xinhao Mei, Chutong Meng, Haohe Liu, Qiuqiang Kong, Tom Ko, Chengqi Zhao, Mark D. Plumbley, Yuexian Zou, and Wenwu Wang. WavCaps: A ChatGPT-Assisted weakly-labelled audio captioning dataset for audio-language multimodal research.

[34] Annamaria Mesaros, Toni Heittola, and Tuomas Virtanen. Tut database for acoustic scene classification and sound event detection. In *2016 24th European Signal Processing Conference (EUSIPCO)*, pages 1128–1132, 2016.

[35] Lars Mescheder, Andreas Geiger, and Sebastian Nowozin. Which training methods for gans do actually converge? In *International conference on machine learning*, pages 3481–3490. PMLR, 2018.

[36] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models.

[37] OpenAI. Gpt-4 technical report, 2023.

[38] William Peebles and Saining Xie. Scalable diffusion models with transformers.

[39] Karol J Piczak. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM International Conference on Multimedia*, pages 1015–1018.

[40] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763.

[41] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, Peter J Liu, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. 21(140):1–67.

[42] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, 2022.

[43] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International Conference on Machine Learning*, pages 8821–8831. PMLR, 2021.

[44] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.

[45] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S Sara Mahdavi, Rapha Gontijo Lopes, et al. Photorealistic text-to-image diffusion models with deep language understanding.

[46] J. Salamon, C. Jacoby, and J. P. Bello. A dataset and taxonomy for urban sound research. In *22nd ACM International Conference on Multimedia (ACM-MM'14)*, pages 1041–1044.

[47] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. Laion-5b: An open large-scale dataset for training next generation image-text models, 2022.

[48] Rico Sennrich, Barry Haddow, and Alexandra Birch. Improving neural machine translation models with monolingual data. *arXiv preprint arXiv:1511.06709*, 2015.

[49] Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. Hugginggpt: Solving ai tasks with chatgpt and its friends in huggingface, 2023.

[50] Uriel Singer, Adam Polyak, Thomas Hayes, Xi Yin, Jie An, Songyang Zhang, Qiyuan Hu, Harry Yang, Oron Ashual, Oran Gafni, et al. Make-a-video: Text-to-video generation without text-video data.

[51] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.

[52] Aaron Van Den Oord, Oriol Vinyals, et al. Neural discrete representation learning. *Advances in neural information processing systems*, 30, 2017.

[53] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[54] Xuenan Xu, Heinrich Dinkel, Mengyue Wu, and Kai Yu. A crnn-gru based reinforcement learning approach to audio captioning. In *DCASE*, pages 225–229, 2020.

[55] Dongchao Yang, Jianwei Yu, Helin Wang, Wen Wang, Chao Weng, Yuexian Zou, and Dong Yu. Diffsound: Discrete Diffusion Model for Text-to-sound Generation.

[56] Daochen Zha, Zaid Pervaiz Bhat, Kwei-Herng Lai, Fan Yang, Zhimeng Jiang, Shaochen Zhong, and Xia Hu. Data-centric artificial intelligence: A survey. *arXiv preprint arXiv:2303.10158*, 2023.

[57] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

Appendices

Make-An-Audio 2: Temporal-Enhanced Text-to-Audio Generation

A Data details

Dataset	Hours	Type	Source
Audiocaps	109hrs	caption	[20]
WavCaps	2056hrs	caption	[33]
WavText5K	25hrs	caption	[3]
MACS	48hrs	caption	[32]
Clothv2	152hrs	caption	[5]
Audiostock	44hrs	caption	https://audiostock.net
epidemic sound	220hrs	caption	https://www.epidemicsound.com
Adobe Audition Sound Effects	26hrs	caption	https://www.adobe.com/products/audition/offers/AdobeAuditionDLCSFX.html
FSD50K	108hrs	label	https://annotator.freesound.org/fsd
ODEON_Sound_Effects	20hrs	label	https://www.paramountmotion.com/odeon-sound-effects
UrbanSound8K	9hrs	label	[46]
ESC-50	3hrs	label	[39]
filteraudioset	945hrs	multi label	[7]
TUT	13hrs	label	[34]

Table 6: Statistics for the Datasets used in the paper.

As shown in Table 6, we collect a large-scale audio-text dataset consisting of 0.92 million of audio samples with a total duration of approximately 3.7k hours. The dataset has a wide variety of sounds including music and musical instruments, sound effects, human voices, nature and living sounds, etc. For Clotho dataset, we only use its evaluation set for zero-shot testing and do not use for training. As speech and music are the dominant classes in AudioSet, we filter 95% of the samples that contain speech and music to build a more balanced dataset.

B Experimental details

Variational autoencoder. We employed a similar VAE architecture to that of Make-An-Audio, replacing all the 2D-convolution layers with 1D-convolution layers and the spatial transformer with a temporal transformer. As detailed in Section 4.5, the output latent of VAE is $z = E(x) \in R^{C_a/f_1 \times T/f_2}$, where we choose the downsample rate of $f_1 = 4$ and $f_2 = 2$. We additionally involve R1 regularization [35] to better stabilize the adversarial training process. We train our VAE on 8 NVIDIA A100 GPU with a batch size of 32 and 800k training steps on AudioSet dataset. We use the Adam optimizer [21] with a learning rate of 1.44×10^{-4}. For specific differences in hyperparameters between our VAE and that of Make-An-Audio, please see Table 7.

Latent diffusion. We train our Latent Diffusion Model with on 8 NVIDIA A100 GPU with a batch size of 32 and 1.2M training steps. We use the Adam optimizer with a learning rate of 9.6×10^{-5}. For the specific hyperparameter for our latent diffusion model, please refer to Table 8.

Model parameters of each component. The params of each component in Make-An-Audio 2 are displayed in Table 9.

	Make-An-Audio VAE	Make-An-Audio 2 VAE
Assume input tensor shape (for 10s audio)	(1,80,624)	(80,624)
Embed_dim	4	20
Convolution layer	Conv2D	Conv1D
Channels	128	224
Channel multiplier	1,2,2,4	1,2,4
Downsample layer position	after block 1,2	after block 1
Attention layer	spatial attention	temporal attention
Attention layer position	after block 3,4	after block 3
Output tensor shape	(4,10,78)	(20,312)

Table 7: Difference between Make-An-Audio VAE and our VAE

	Make-An-Audio 2 LDM
Input shape	(20,T)
Condition_embedding dim	1024
Feed-forward Transformer hidden_size	576
Feed-forward Transformer's Conv1d kernel size	7
Feed-forward Transformer's Conv1d padding	3
Number of Transformer heads	8
Number of Feed-forward Transformer block	8
Diffusion steps	1000

Table 8: Make-An-Audio 2 Diffusion model backbone configurations

C Evaluation

C.1 Subjective evaluation

To assess the generation quality, we conduct MOS (Mean Opinion Score) tests regarding audio quality and text-audio faithfulness, respectively scoring MOS-Q and MOS-F.

For audio quality, the raters were explicitly instructed to "focus on examining the audio quality and naturalness." The testers were presented with audio samples and asked to rate their subjective score (MOS-P) on a 20-100 Likert scale.

For text-audio faithfulness, human raters were shown the audio and its caption and asked to respond to the question, "Does the natural language description align with the audio faithfully?" They had to choose one of the options - "completely," "mostly," or "somewhat" on a 20-100 Likert scale.

Our crowd-sourced subjective evaluation tests were conducted via Amazon Mechanical Turk where participants were paid $8 hourly. A small subset of the generated audio samples used in the test can be found at `https://make-an-audio-2.github.io/`.

Component	Params
VAE	213M
Diffusion Model Backbone	160M
Text Encoder	452M
Vocoder	112M
Total	937M

Table 9: The params of each component in Make-An-Audio2

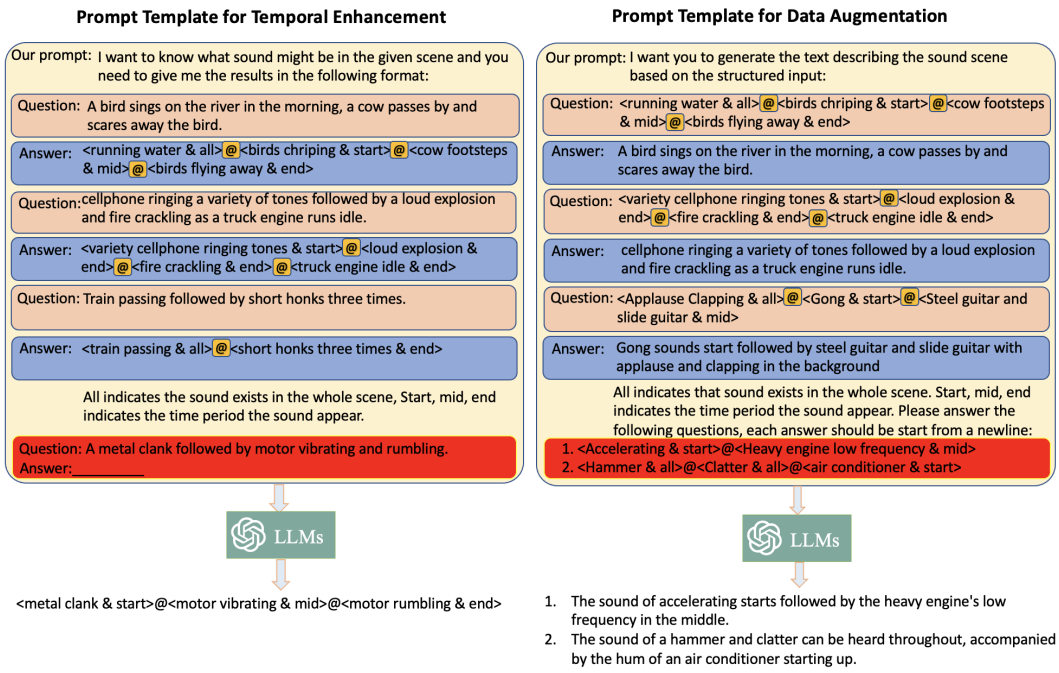

Figure 3: The prompt templates we used for temporal enhancement and data augmentation. We use the symbol '&' to split the sound event and the time order. We use the symbol '@' to split <event & order> pairs.

C.2 Objective evaluation

Fréchet Audio Distance (FAD) [19] is adapted from the Fréchet Inception Distance (FID) to the audio domain, it is a reference-free perceptual metric that measures the distance between the generated and ground truth audio distributions. FAD is used to evaluate the quality of generated audio.

The inception Score (IS) is an effective metric that evaluates both the quality and diversity of generated audio.

KL divergence is measured at a paired sample level between the generated audio and the ground truth audio, it is computed using the label distribution and is averaged as the final result.

Fréchet Distance (FD) evaluates the similarity between the generated and ground truth audio distributions. FD, KL and IS are built upon an audio classifier, PANNs [24], which takes the mel-spectrogram as model input. Differently, FAD uses VGGish [10] as an audio classifier that takes raw audio waveform as model input.

CLAP score: adapted from the CLIP score [11, 40] to the audio domain and is a reference-free evaluation metric to measure audio-text alignment for this work that closely correlates with human perception.

D ChatGPT prompts

The prompt templates utilized for temporal enhancement to construct structure caption from the original natural language caption and for caption data augmentation are displayed in Figure 3.

Table 10 presents some instances of the original caption and ChatGPT's outcome. For text data augmentation, we construct structured caption inputs, and Table 11 exhibits examples of such inputs and ChatGPT's corresponding outputs.

Natural language input	ChatGPT's output
A woman talks nearby as water pours	<woman talking& all>@<water pouring& all>
Two men exchange words, then a car engine revs followed by a siren and fade in music	<two men talking& all>@<car engine revving& start>@<siren& mid>@<music fading in& end>
A crowd is cheering and shouting, thumping occurs, an adult female speaks, and an adult male speaks	<crowd cheering and shouting& all>@<thumping& start>@<adult female speaking& mid>@<adult male speaking& end>

Table 10: Examples of using ChatGPT for temporal enhancement from AudioCaps trainset

Structured input	ChatGPT's output
<bark howl dog& start>@<typing Typewriter& mid>@<breathing& end>	A dog barks and howls while someone types on a typewriter, then the sound of breathing takes over
<crowing& all>@<car& all>@<female singing& end>	A rooster crows as a car passes by, and the scene ends with the sound of a female singing
<sneezing& all>@<bicycle bell ring& start>@<typewriter & end>	The sound of sneezing is heard throughout, with a bicycle bell ringing at the start and the sound of a typewriter at the end

Table 11: Examples of using ChatGPT for data augmentation

E Limitations, future works and broader impact

Limitations. Make-An-Audio 2 incorporates an additional LLM for parsing the original caption, which affects both the generation performance and running speed. Additionally, the generative diffusion model employed by Make-An-Audio 2 requires multiple iterative refinements for synthesis, which can be time-consuming to produce high-quality results. Furthermore, the speech in the generated audio can be intelligible.

Future works. We leave the T2A system which supports speech synthesis for future work. In addition, we aim to implement T2A systems that could take structured inputs as optional auxiliary inputs instead of required inputs.

Broader impacts. At the same time, we acknowledge that Make-An-Audio 2 may lead to unintended consequences such as increased unemployment for individuals in related fields such as sound engineering and radio hosting. Furthermore, there are potential concerns regarding the ethics of non-consensual voice cloning or the creation of fake media.